

④ classify the square Matrices as singular or non-singular given by

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 2 & -2 & 1 \\ 2 & 2 & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

→ Now in this case what we will do?
we will find the determinant ~~for~~^{of} both Matrices.
first we will find the determinant of Matrix A.
and then B.

then we will get $|A|$ of A is 0 means
A is singular Matrix.

same for Matrix B.

then we will get determinant of B we will get

$$|B| = -12$$

Hence it is $|B|$ is Non singular.